Aurora: Contrastive Residual Forecasting for Industrial Time-Series Anomaly Detection

Synthetic excerpt prepared for the LabRelay demo corpus. Not a real publication.

Research question: detect rare anomalies in multivariate industrial sensor streams

without per-machine labels. The claim is that a residual forecaster trained with a

contrastive objective matches supervised detectors while needing only normal data.

Core method: a one-step forecaster predicts the next window; the anomaly score is the

normalized residual energy, and the contrastive term pushes normal residuals together

and rare anomalous residuals apart. No closed-form bound is claimed.

Experimental setup: 24000 windows, 80/20 train-test split, AdamW optimizer, learning

rate 3e-4, batch size 128, 40 epochs, random seed 2026, data version v3.

Results: AUROC 0.912 and F1 0.845 on the held-out test split. Reproduction reaches the

same numbers when the feature cache directory is empty.

Known limitation: the ablation over random seeds was never completed, so the variance

of the reported AUROC is unknown. A stale feature cache drops AUROC to 0.874 and is the

first thing to rule out before suspecting the model.

Conclusion: the pipeline is reproducible from the configuration alone; the unfinished

work is the seed ablation followed by a learning-rate sweep.